

Week 1: From Scripts to Systems

Understand what makes an AI system different from a plain script, grasp the Orchestrator and Sub-Agent architecture, and build the skeleton of the iterative agentic loop.

MLE-STAR ARCHITECTURE

WEEK 1 OF 4



by **Alwin Lin**

Today's Session

1 — Course At A Glance

What are we teaching

2 — Course Syllabus

What are we looking for

3 — Welcome & End Goal

What is MLE-STAR

4 — Infrastructure & API Setup

Wrappers Wrappers

5 — Q&A

Clarify Issues before Week 2

Course At A Glance

Sunday 08:00~09:00 (GMT +8) / Saturday 17:00~18:00 (GMT -8), notify 3 days prior to class

1. **Week 1 - 04/19**

a. The Agentic Loop

2. **Week 2 - 04/26**

a. Tool Use & Context

3. **Week 3 - 05/03**

a. System Architecture

4. **Week 4 - 05/10**

a. Milestone 1: Coder

1. **Week 5 - 05/17 [Office Hour 1]**

a. Milestone 2: Benchmark

2. **Week 6 - 05/24 [Coaching session 1
Rehearsal 1]**

a. Milestone 3: Refiner

3. **Week 7 - 05/31[Office Hour 2]**

a. Milestone 4: Integration

4. **Week 8 - 06/07[Coaching and Rehearsal 2]**

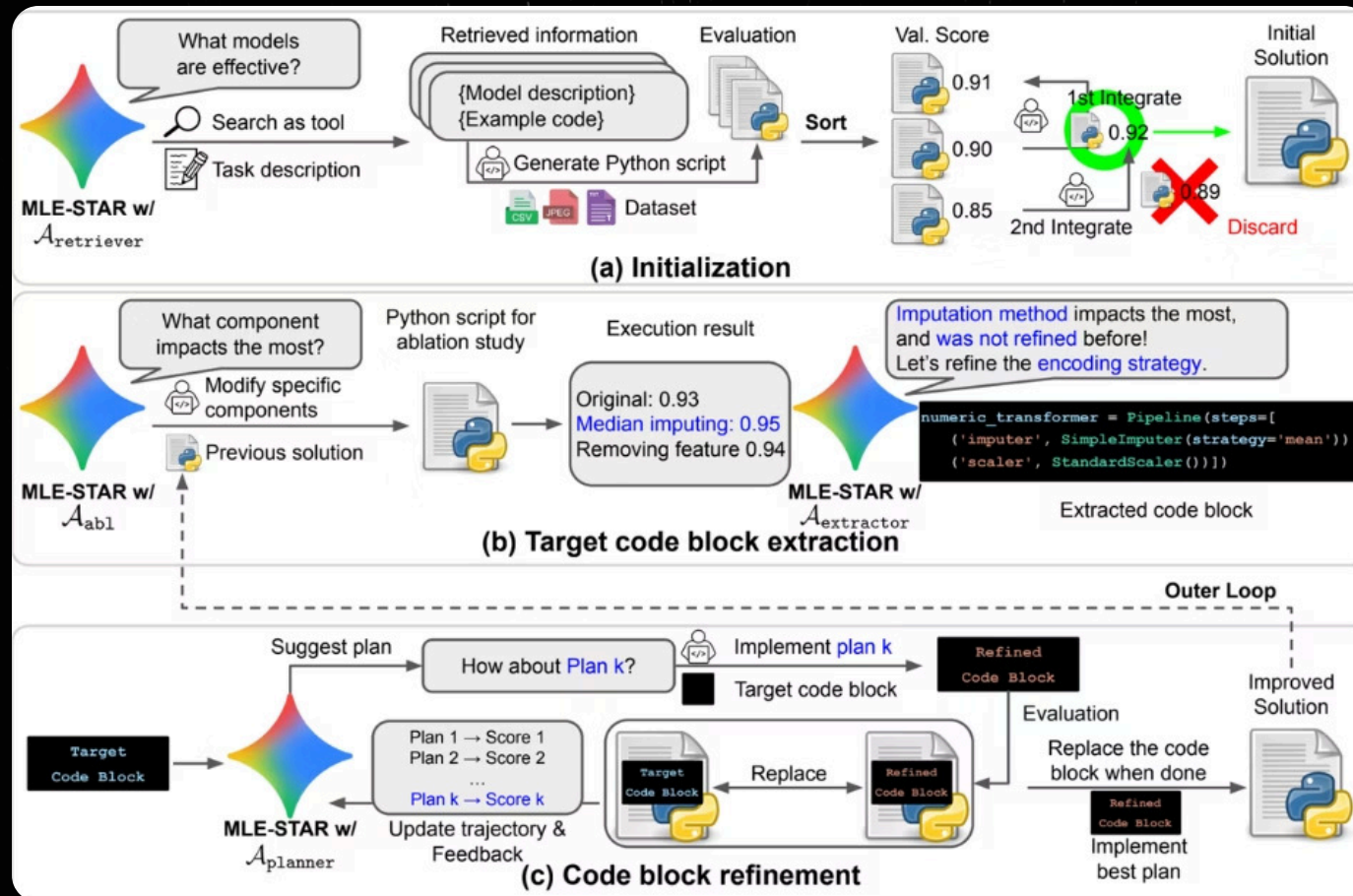
a. Demos Rehearsal

Grading Rubric

- System Architecture & Components (20 Points)
 - Whether all required modules were built and structured correctly
- The Autonomous Loop (30 Points)
 - Evaluates the core engineering challenge: the `while` loop
- Task Correctness & Benchmarking (20 Points)
 - How well the agent's generated code actually performs
- Robustness & API Safety (15 Points)
 - Safety constraints and edge-case handling
- Presentation & Code Clarity (15 Points)
 - Code clarity and communication



Welcome to MLE-Star



Welcome to MLE-Star

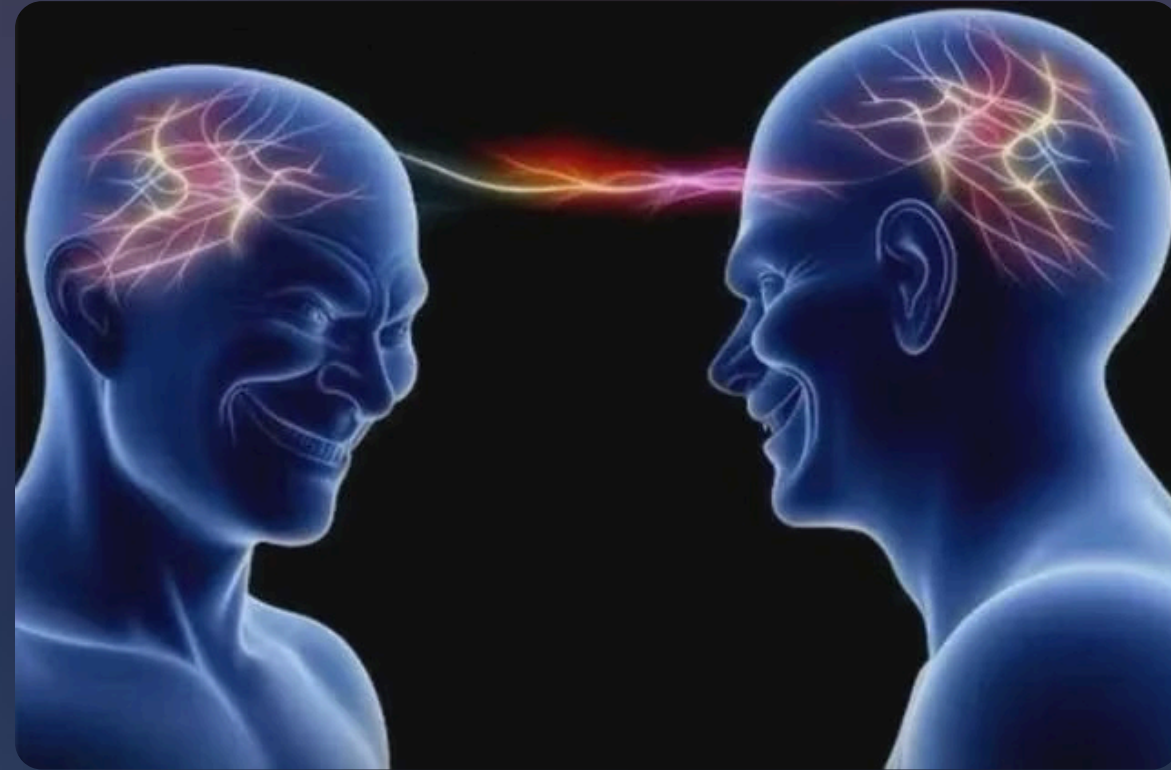
MLE-Star

What is it

What Does it Do

What is the significance

Why all of a sudden



MLE-Star

What is it

What Does it Do

What is the significance

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What is the significance

Model	Made Submission (%)	Valid Submission (%)	Above Median (%)	Bronze (%)	Silver (%)	Gold (%)	Any Medal (%)
MLE-STAR (Ours)							
gemini-2.5-pro	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	83.3 $\pm$ 4.6	6.1 $\pm$ 3.0	21.2 $\pm$ 5.1	36.4 $\pm$ 6.0	63.6 $\pm$ 6.0
gemini-2.0-flash	95.5 $\pm$ 2.6	95.5 $\pm$ 2.6	63.6 $\pm$ 6.0	9.1 $\pm$ 3.6	4.5 $\pm$ 2.6	30.3 $\pm$ 5.7	43.9 $\pm$ 6.2
AIDE (Jiang et al., 2025)							
gemini-2.0-flash	87.9 $\pm$ 4.0	78.8 $\pm$ 5.0	39.4 $\pm$ 6.0	4.5 $\pm$ 2.6	9.1 $\pm$ 3.5	12.1 $\pm$ 4.0	25.8 $\pm$ 5.4
o1-preview	99.7 $\pm$ 0.3	90.3 $\pm$ 1.6	58.2 $\pm$ 2.6	4.8 $\pm$ 1.1	11.1 $\pm$ 1.7	20.7 $\pm$ 2.2	36.6 $\pm$ 2.6
gpt-4o	82.1 $\pm$ 1.4	65.7 $\pm$ 1.7	29.9 $\pm$ 1.6	3.4 $\pm$ 0.6	5.8 $\pm$ 0.8	9.3 $\pm$ 1.0	18.6 $\pm$ 1.4
llama-3.1-405b-instruct	72.7 $\pm$ 5.5	51.5 $\pm$ 6.2	18.2 $\pm$ 4.7	0.0 $\pm$ 0.0	4.5 $\pm$ 2.6	6.1 $\pm$ 2.9	10.6 $\pm$ 3.8
claude-3-5-sonnet	81.8 $\pm$ 4.7	66.7 $\pm$ 5.8	33.3 $\pm$ 5.8	3.0 $\pm$ 2.1	6.1 $\pm$ 2.9	10.6 $\pm$ 3.8	19.7 $\pm$ 4.9
MLAB (Huang et al., 2024a)							
gpt-4o	84.8 $\pm$ 4.4	63.6 $\pm$ 5.9	7.6 $\pm$ 3.3	3.0 $\pm$ 2.1	1.5 $\pm$ 1.5	1.5 $\pm$ 1.5	6.1 $\pm$ 2.9
OpenHands (Wang et al., 2024)							
gpt-4o	81.8 $\pm$ 4.7	71.2 $\pm$ 5.6	16.7 $\pm$ 4.6	3.0 $\pm$ 2.1	3.0 $\pm$ 2.1	6.1 $\pm$ 2.9	12.1 $\pm$ 4.0

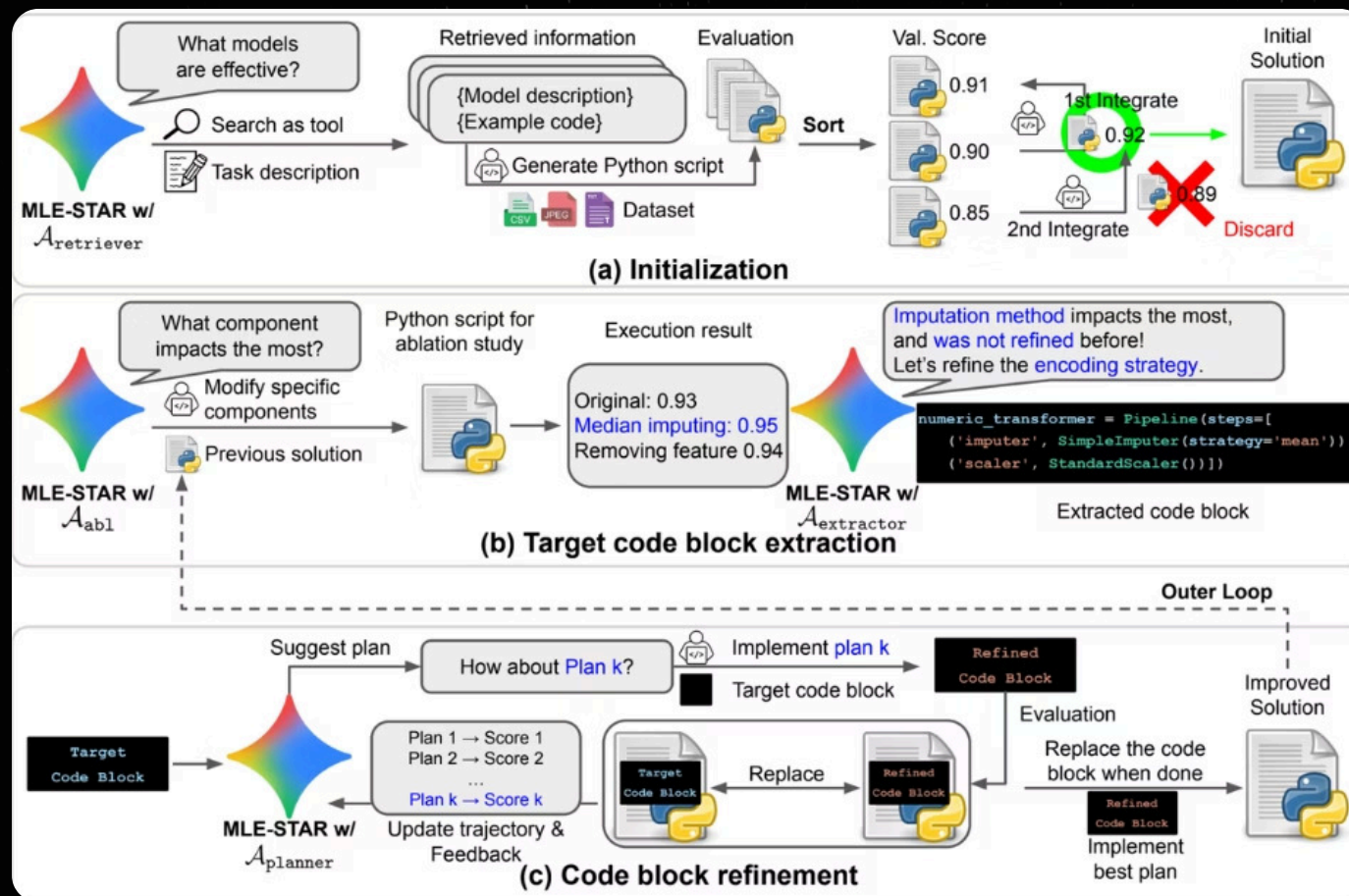
Cool, but at what cost

F. Experimental setup

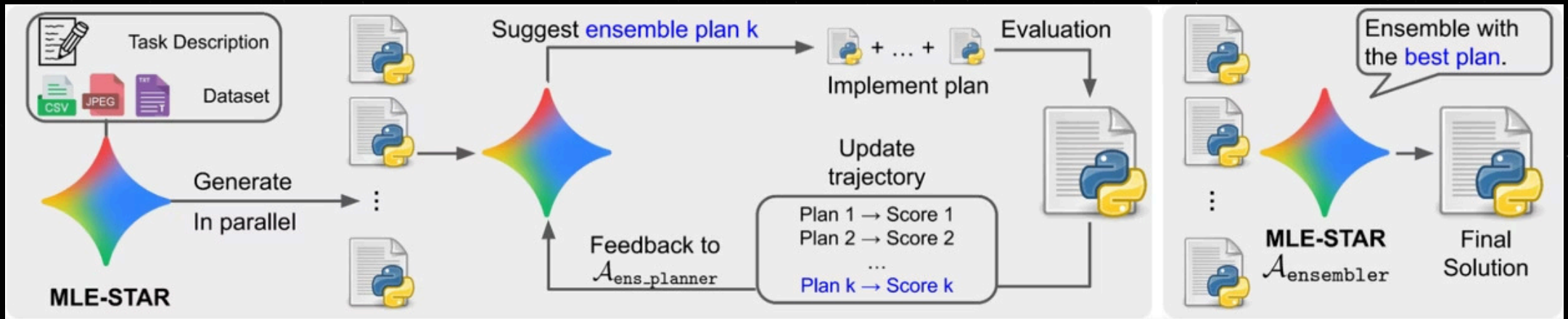
We conducted our experiments mainly using 96 vCPUs with 360 GB Memory (Intel(R) Xeon(R) CPU), and 8 NVIDIA V100 GPUs with 16 GB Memory.

Required time to generate a single solution using MLE-STAR. With the configuration of four retrieved models, four inner loops, four outer loops, and five rounds for exploring the ensemble strategy, MLE-STAR requires 14.1 hours to generate a single final solution, on average across 22 tasks and all three random trials (*i.e.*, total 66 experiments). On the other hand, we found that AIDE ([Jiang et al., 2025](#)) requires 15.4 hours. This indicates that our method does not require more time to run compare to the best alternative. Note that a maximum time limit of 24 hours was set for both methods, following the MLE-bench's experimental setup.

MLE-Star - Core Concept



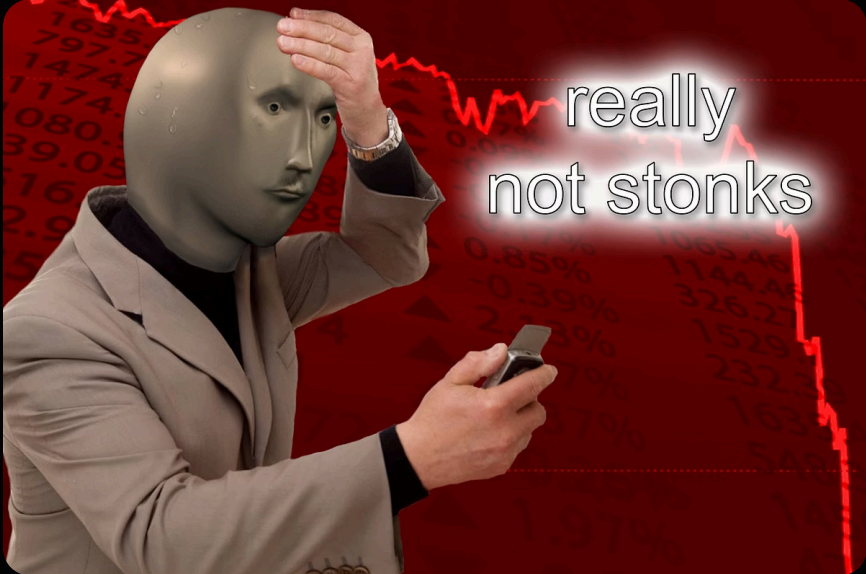
TLDR:



Why Many Agent When One Good?

Subagents and loops:

- Why do they even exists?
- What are the use case?



really
not stonks

Some Very Bad News

"As of early 2026, Google has significantly altered the Gemini API free tier to curb abuse, **lowering rate limits by 50-80%**"

- Gemini 3.1 Flash Lite
- Gemini 2.5 Flash Lite
- Gemini 3 Flash



Somewhat Good news

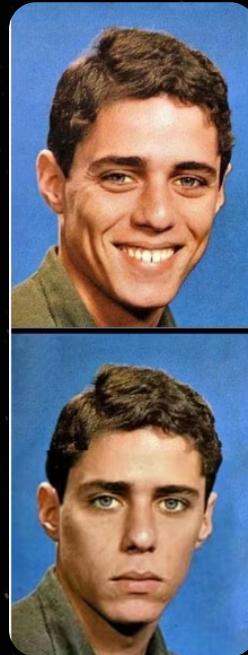
"Google released Gemma 4 on **April 2, 2026**"

- Gemma 4 26B | 15 RPM | Unlimited TPM | 1.5K RPD
- Gemma 4 31B | 15 RPM | Unlimited TPM | 1.5K RPD

What Does It look Like?

What does your second project look like?

- User Input
- Search triggered
- Return top "X" result
- Close



How can we make it autonomous?

- Three agents
 - Director
 - Manager
 - Researcher

Q&A?

